

AdTEC: A Unified Benchmark for Evaluating Text Quality in Search Engine Advertising

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CyberAgent AI Lab NAIST Presented at NAACL 2025

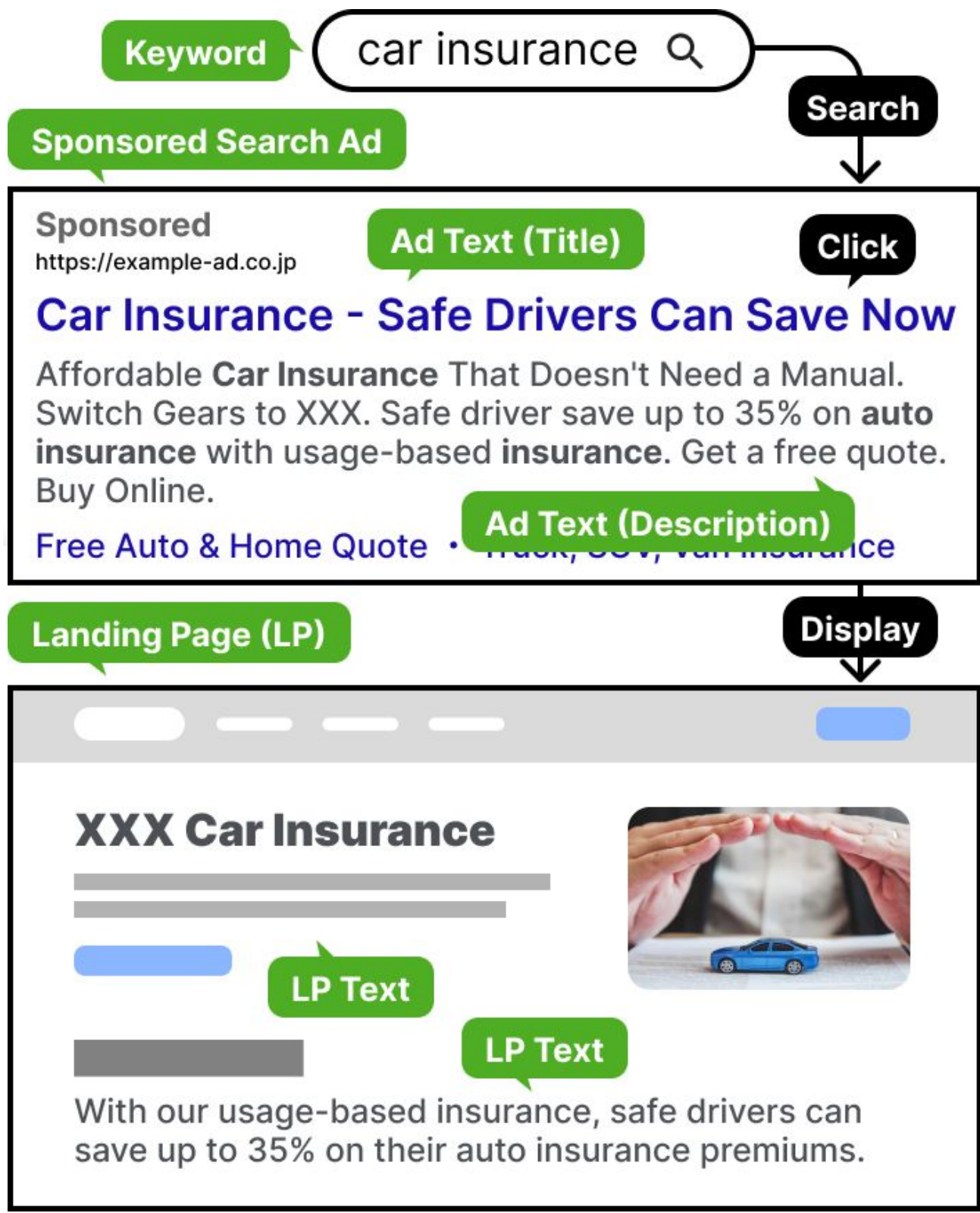


AdTEC (Ad Text Evaluation Benchmark by CyberAgent)

is **the first public dataset** designed for **evaluating the quality of ad texts** based on **real-world AdOps workflows**.

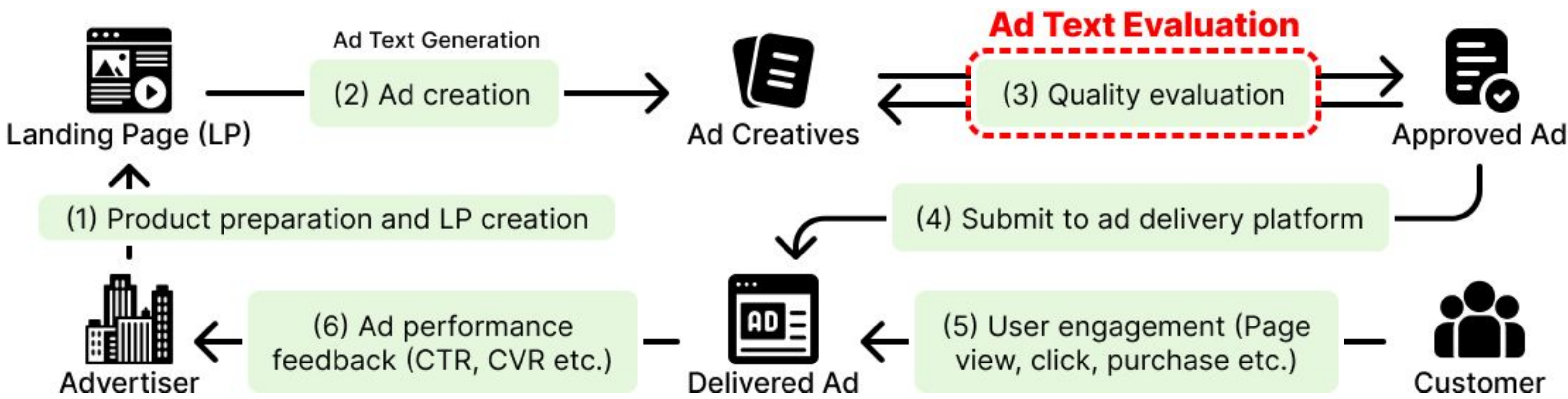
- Online advertising is a huge industrial use case for NLP
 - Worldwide digital advertising spending is expected to reach \$723B^[1] in 2026 and NLP (GenAI) can generate ad texts (e.g. Sponsored Search Ads) massively. [2, 3, 4, 5]
- How can we ensure the quality of mass-produced ad creatives?
 - Organizing the workflow of advertising operations (AdOps)
 - Creating a benchmark to evaluate ad creatives

An Example of Sponsored Search Ad



Dataset Construction

We interviewed experts in the advertising agencies and organized the following AdOps workflow.



We then identify crucial parts within Step 3 and generalized and defined tasks that adhere to the principle of evaluating ad texts either *directly* or *indirectly*.

- AD ACCEPTABILITY** [13265/970/980] Predicting the acceptance of overall quality
- AD CONSISTENCY** [10635/945/970] Predicting the consistency between LP content and ad text
- AD PERFORMANCE ESTIMATION** [125087/965/965] Estimating the quality score with the information of ad delivery
- A3 (ASPECT of AD APPEAL) RECOGNITION** [1856/465/410] Recognizing the A3s in the ad text [6]
- AD SIMILARITY** [4980/623/629] Predicting the similarity of a pair of ad texts

Task	Setup: Input → Label	Metrics
AD ACCEPTABILITY	Classification: Ad text → acceptable/unacceptable	Accuracy/F1-score
AD CONSISTENCY	Classification: (Ad text, LP text) → consistent/inconsistent	Accuracy/F1-score
AD PERF. EST.	Regression: (Ad texts, Keyword, Industry) → [0, 100]	Pearson/Spearman corr.
A ³ RECOGNITION	Classification: Ad text → multi-labels (See Appendix Table 11)	F1-micro/-macro
AD SIMILARITY	Regression: Ad text pair → [1, 5]	Pearson/Spearman corr.

Data Collection

- Actual AdOps workflow in CyberAgent
- Delivered ads between 2021 and 2022
- All data are in Japanese

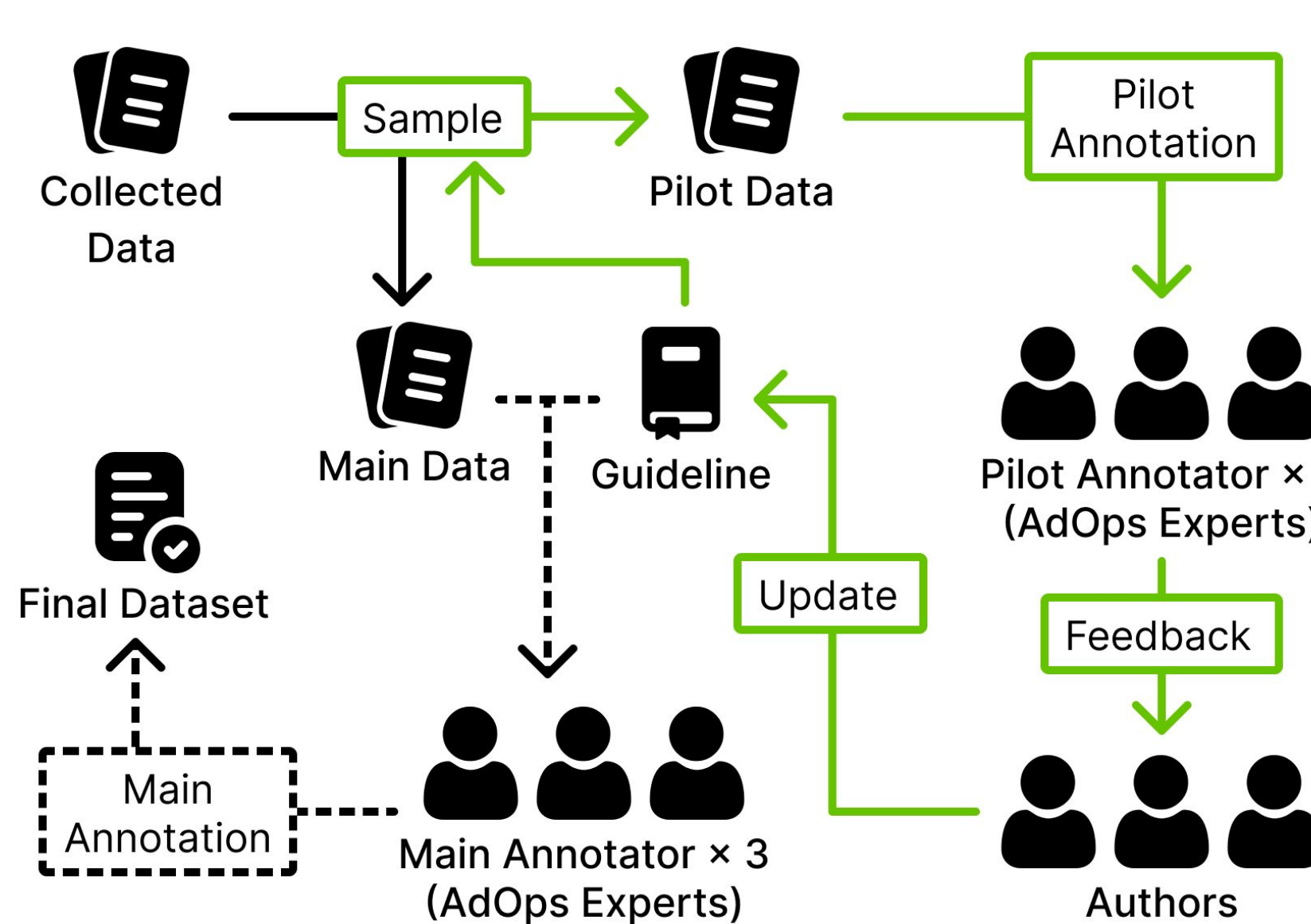
Data Pre-processing

- Deduplication
- Masking the sensitive entities
- Pseudo-similar pairs based on the ad hierarchical structure (*)

Data Splitting

- Based on the ad hierarchical structure (*)

Annotation Workflow



Green line: Guideline Creation (repeated twice)
Dashed line: Main Annotation (only once)

Experiments

We evaluated the performance on our AdTEC benchmark using PLMs and human evaluators.

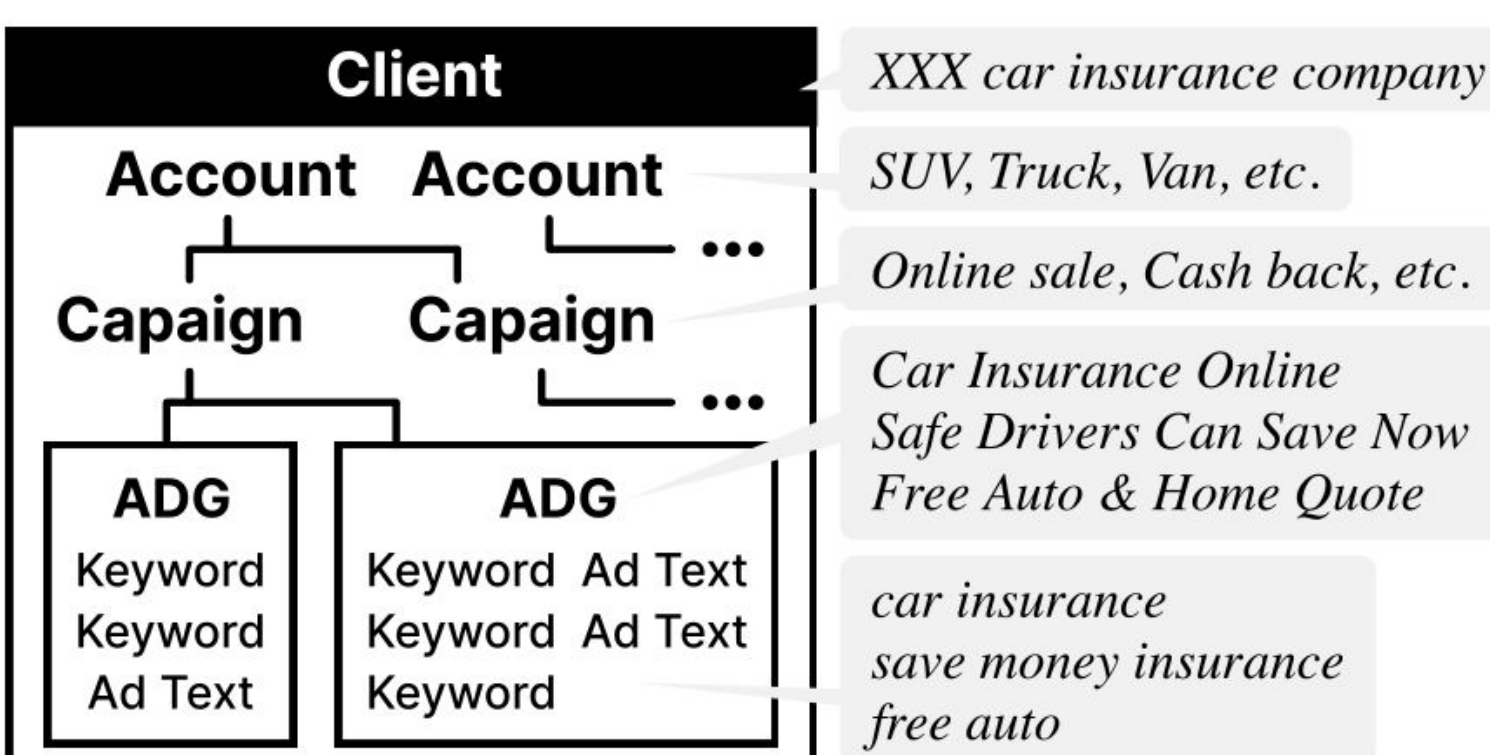
Evaluator	AD ACCEPT. Accuracy/F1-score	AD CONSIST. Accuracy/F1-score	AD PERF. EST. Pearson/Spearman	A ³ RECOGNITION F1-micro/-macro	AD SIMILARITY Pearson/Spearman
Fine-tuned Encoder Models					
Tohoku BERT	0.711/0.688	0.767/0.552	0.480*/0.497*	0.774/0.694*	0.773/0.807
Waseda BERT	0.598/0.637	0.755/0.474	0.445/0.457	0.663/0.517	0.740/0.800
XLM-RoBERTa	0.705/0.690	0.758/0.519	0.453/0.457	0.778/0.677	0.878*/0.878*
Zero-/Few-shot LLMs					
CALM2 _{7b}	0.520/0.115	0.381/0.472	0.006/0.013	0.154/0.042	0.036/0.036
ELYZA _{7b}	0.352/0.520	0.628/0.771	0.003/0.046	0.196/0.044	0.015/-0.004
GPT-4	0.325/0.433	0.583/0.612	0.028/0.073	0.417/0.113	0.776/0.811
Human	0.732*/0.790*	0.703/0.807*	—	0.564/0.538	0.699/0.765

- Fine-tuning encoder models still perform well in AdTEC
- Humans surpassed PLMs in AD ACCEPTABILITY and CONSISTENCY tasks
 - In AD ACCEPTABILITY, models miss semantic issues such as redundancy or contradiction
 - In AD CONSISTENCY, models struggled when LPs omit expressions like 'official'.
- GPT-4 and fine-tuned models beat human performance in AD SIMILARITY
 - PLMs can better understand nuanced relationships between ad content
- Open-source LLMs struggled with semantic similarity and numerical prediction tasks

True Label	Pred. Label	Legend: ■ Negative □ Neutral ■ Positive
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- PLMs focus on surface features and miss semantic similarity and paraphrases
- Enhancing performance needs better handling of meta-expressions and reasoning

(*) Ad Hierarchical Structure



References

- PwC, Changes in digital advertising spend, 2025
- Hughes et al., Generating Better Search Engine Text Advertisements with Deep Reinforcement Learning, 2019
- Kamigaito et al., An Empirical Study of Generating Texts for Search Engine Advertising, 2021
- Golobokov et al., DeepGen: Diverse Search Ad Generation and Real-Time Customization, 2022
- Mita et al., Striking Gold in Advertising: Standardization and Exploration of Ad Text Generation, 2024
- Murakami et al., Aspect-based Analysis of Advertising Appeals for Search Engine Advertising, 2022